1. Wazuh

Task 1 - Introduction

Wazuh is a free and open-source security platform that provides threat detection, integrity monitoring, incident response, and compliance capabilities. Wazuh can collect data from various sources, including logs, events, network traffic, and cloud metadata. It can automate response actions, such as blocking malicious traffic or isolating infected hosts.

Wazuh operates on a manager and agent model. Simply put, there is one Wazuh server (**manager**) that stores and processes the data, and many hosts (**agents**) that send data to the manager.

Answer the questions below

> When was Wazuh released? 2015

> What is the term that Wazuh calls a device that is being monitored for suspicious activity and potential security threats? Agent

> Lastly, what is the term for a device that is responsible for managing these devices? Manager

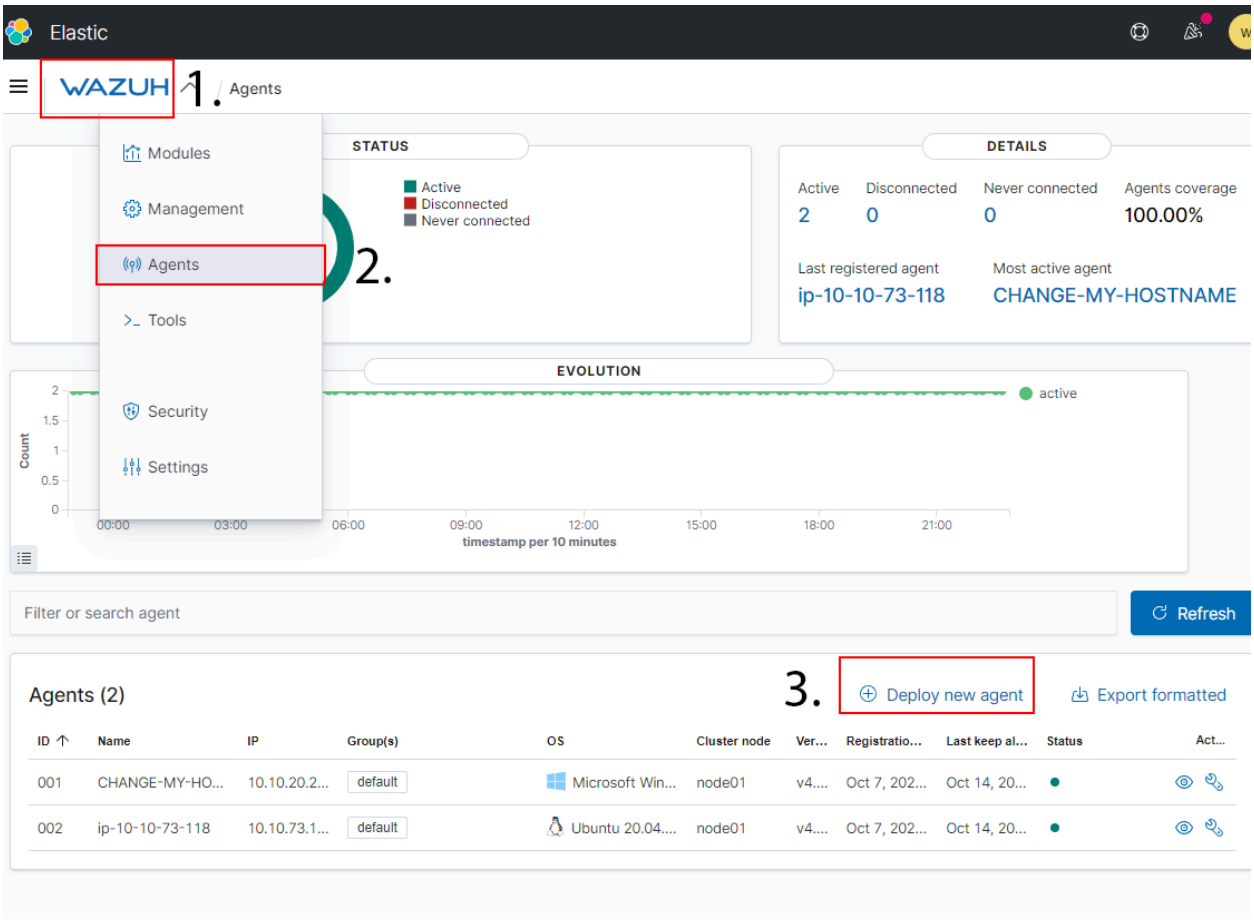
Task 2 - Deploy Wazuh Server

To Deploy or to install: Wazuh - Open Source XDR. Open Source SIEM.

Task 3 - Wazuh Agents

Devices that record the events and processes of a system are called agents. Agents monitor the processes and events that take place on the device, such as authentication and user management. Agents will offload these logs to a designated collector for processing, such as Wazuh.

This wizard can be launched by navigating to the following location on the Wazuh server: **Wazuh → Agents → Deploy New Agent** as illustrated in this screenshot below:



Answer the questions below

> How many agents does this Wazuh management server manage? 2

> What are the status of the agents managed by this Wazuh management server? Disconnected

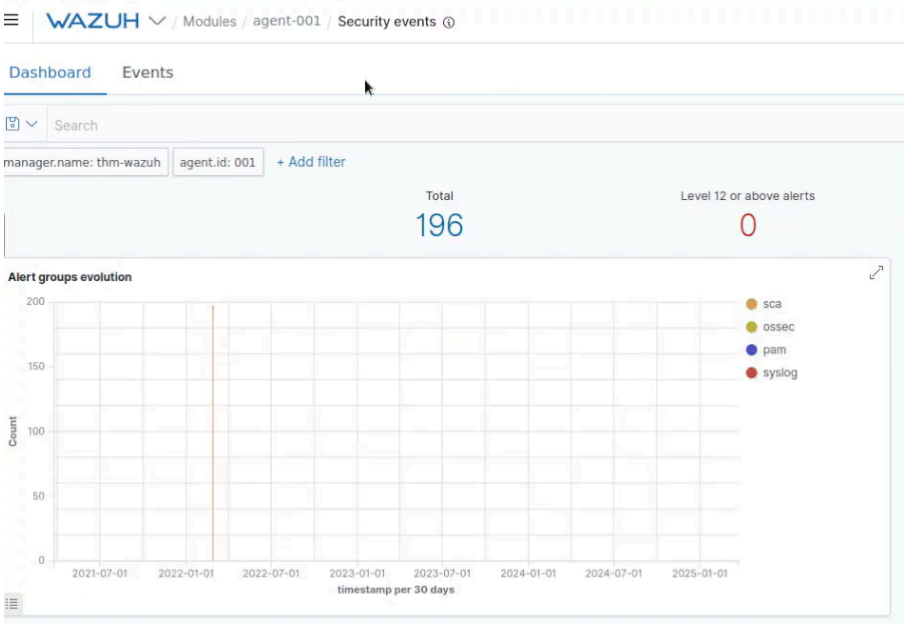
Task 4 - Wazuh Vulnerability Assessment & Security Events

Wazuh's Vulnerability Assessment module is a powerful tool that can be used to periodically scan an agent's operating system for installed applications and their version numbers. Once this information has been gathered, it is sent back to the Wazuh server and compared against a database of CVEs to discover potential vulnerabilities.

Answer the questions below

> How many "Security Event" alerts have been generated by the agent "AGENT-001"?

Note: You will need to make sure that your time range includes the 11th of March 2022 196



Task 5 - Wazuh Policy Auditing

Wazuh is capable of auditing and monitoring an agent's configuration whilst proactively recording event logs. When the Wazuh agent is installed, an audit is performed where a metric is given using multiple frameworks and legislations such as NIST, MITRE and GDPR.

Task 6 - Monitoring Logons with Wazuh

Wazuh's security event monitor is capable to actively record both successful and unsuccessful authentication attempts. The rule with an id of 5710 detects attempted connections that are unsuccessful for the SSH protocol.

Task 7 - Collecting Windows Logs with Wazuh

All sorts of actions and events are captured and recorded on a Windows operating system. This includes authentication attempts, networking connections, files that were accessed, and the behaviours of applications and services. This information is stored in the Windows event log using a tool called Sysmon. We can use the Wazuh agent to aggregate these events recorded by *Sysmon* for processing to the Wazuh manager. Now, we will need to configure both the Wazuh agent and the Sysmon application. Sysmon uses rules that are made in XML formatting to be triggered. To instruct Sysmon to do, we need to execute the Sysmon application and provide the aforementioned configuration file like so: `Sysmon64.exe -accepteula -i detect_powershell.xml`

Answer the questions below

> What is the name of the tool that we can use to monitor system events? Sysmon

> What standard application on Windows do these system events get recorded to? Event Viewer

Task 8 - Collecting Linux Logs with Wazuh

Capturing logs from a Linux agent is a simple process similar to capturing events from a Windows agent. We will be using Wazuh's log collector service to create an entry on the agent to instruct what logs should be sent to the Wazuh management server. Wazuh comes with many rules that enable Wazuh to analyze log files and can be found in `/var/ossec/ruleset/rules`. Some common applications include:

- Docker
- FTP
- WordPress
- SQL Server
- MongoDB
- Firewalld
- And many, many more (approximately 900).

However, you can always make your own rules. In this task, Wazuh will digest *Apache2* logs using the `0250-apache_rules.xml` ruleset.

Answer the questions below

> What is the full file path to the rules located on a Wazuh management server?

/var/ossec/ruleset/rules

Task 9 - Auditing Commands on Linux with Wazuh

Wazuh utilises the `auditd` package that can be installed on Wazuh agents running on Debian/Ubuntu and CentOS operating systems. In this task, we will be using `auditd` on a Ubuntu system. `Auditd` monitors the system for certain actions and events and will write this to a log file.

Answer the questions below

> What application do we use on Linux to monitor events such as command execution? auditd

> What is the full path & filename for where the aforementioned application stores rules? /etc/audit/rules.d/audit.rules

Task 10 - Wazuh API

Using Our Own Client: The Wazuh management server features a rich and extensive API to allow the Wazuh management server to be interacted with using the command line. Because the Wazuh management server requires authentication, we must first authenticate our client.

Using Wazuh's API Console: Wazuh has a powerful, integrated API console within the Wazuh website to query management servers and agents. Whilst it is not as extensive as using your own environment (where you can create and run scripts using python, for example), it is convenient.

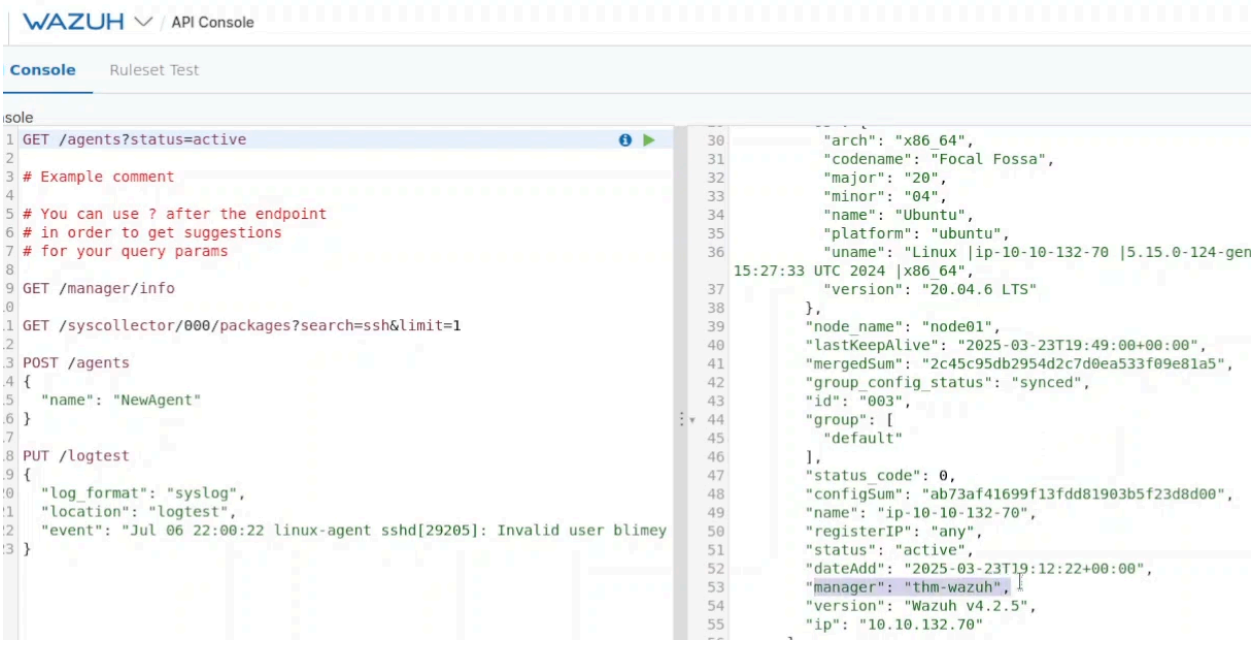
Answer the questions below

> What is the name of the standard Linux tool that we can use to make requests to the Wazuh management server? curl

> What HTTP method would we use to retrieve information for a Wazuh management server API? GET

> What HTTP method would we use to perform an action on a Wazuh management server API? PUT

> Use the API console to find the Wazuh server's version.Note: You will need to add the "v" prefix to the number for this answer. v4.2.5



Task 11 - Generating Reports with Wazuh

Wazuh features a reporting module that allows you to view a summarised breakdown of events that have occurred on an agent. To do so, I will need to open the view: **1. Modules → 2. Security Events.** The report **may** take between a couple of seconds to a few minutes to generate (depending on the amount of data needed to be processed). After allowing some time, we will navigate to the report overview dashboard within Wazuh.

> Analyse the report. What is the name of the agent that has generated the most alerts? agent-001

Task 12 - Loading Sample Data

The Wazuh management server comes with sample data bundled with the installation that can be loaded at your convenience. I have not enabled this by

default to improve the performance of the server. However, if you wish to import much more data to showcase the extensibility of Wazuh further, follow the steps below. Navigate to the module to load the sample data:

1. Open the "**Wazuh"** tab in the heading.
2. Highlight over "**Settings"**.
3. Select the "**Sample Data"** heading.
4. Press the "**Add Data**" button on the respective three cards to import the data.